

UI NOTIFICATIONS SYSTEM

OVERVIEW

The UI Notifications System oversees most of the alerts, primarily errors, that the player sees. For example, it displays the “Too Much Carbon” and “Not Enough People” alerts that appear when a tile can’t be placed. The primary script behind this system is `unableToPlaceTileUI.cs`. It sits on the “Notification System” game object, which is a child of the “Error Displays Canvas” (Under the “UI Canvases” game object). Also involved in this system is the `ErrorGraphicDisplayer.cs` script. Please note that the systems that display tool tips over tiles’ buttons and yellow tutorial tips in the upper left are separate systems. For more information on them, see “[Tool_Tip_System.pdf](#)” and “[Tutorial_Tip_System.pdf](#)”.

HOW THE SYSTEM WORKS

For each notification that can appear on the player's screen, there's a game object that holds the notification's graphic. Here's what some of them look like:



Each of these game objects is stored in an array called “errorGraphics” in the `unableToPlaceTileUI` class. At the beginning of the game, the `unableToPlaceTileUI` class creates an instance of the `ErrorGraphicDisplay` class for each error graphic and then stores all of them in an array called “errorGraphicTimers”. Each error graphic displayer can make its graphic appear and do animations.

The `unableToPlaceTileUI` class has a public function to display each notification. Whenever a script wants an error notification to appear, it will use `unableToPlaceTileUI`'s singleton and call the function corresponding to the correct notification, like this: “`unableToPlaceTileUI._unableToPlaceTileUI.tooMuchCarbon()`”. Then, the `unableToPlaceTileUI` singleton will automatically call the function “`DisplayErrorGraphic`” for the corresponding `ErrorGraphicDisplay` class, like this: “`errorGraphicTimers[4].DisplayErrorGraphic()`”.

The error graphic displayer will display the graphic for a specified amount of time, using the “`FadeGraphic`” and “`ShakeGraphic`” scripts attached to each error graphic game object to emphasize the notification.

HOW TO ADD A NEW NOTIFICATION

This section describes how to create a new type of notification. Sorry the process is a little convoluted. If you have time, feel free to clean it up. If I were to rewrite this system now, I would make it events based (like the audio system) so that scripts don't have to make calls to the `unableToPlaceTileUI` singleton. (Among other things, I would also give the class a better name.)

1. Create the notification's game object.

- a. Locate all the other notification game objects in the game object hierarchy. They're located under "UI Canvases" > "Error Displays Canvas" > "Notification System".
- b. Copy and paste one of the other game objects, replacing its graphics with the new notification's graphics.

2. Add it to the `errorGraphics` array:

In the inspector, on the "Notification System" game object, under the `unableToPlaceTileUI` game object, add a new entry to the "Error Graphics" array. Drag the new notification's game object into the blank slot.

3. Add how long it should be shown to the "Times To Show Graphics" array.

4. Create its function:

In the `unableToPlaceTileUI.cs` script, add a new function that will display the notification. It should look like this, replacing "NewExampleNotification" with the notification's name and the "X" with the game object's position in the `errorGraphics` array:

```
public void NewExampleNotification(){  
    errorGraphicTimers[X].DisplayErrorGraphic();  
}
```

5. Call the new function:

In whatever script you want to call the notification, use this function:

```
unableToPlaceTileUI._unableToPlaceTileUI.ExampleNotification()
```

INTERNAL API

unableToPlaceTileUI.cs

The `unableToPlaceTileUI` class is accessible with the singleton `unableToPlaceTileUI._unableToPlaceTileUI` (Sorry about the weird notation.)

`void tooMuchCarbon()`

Displays the graphic that says “Too Much Carbon!” when the player tries to place a tile that produces carbon when the carbon levels are already maxed out.

`void notEnoughMoney()`

Displays the graphic that says “Not Enough Money” when the player tries to place a tile they don’t have enough money for.

`void notEnoughHomes()`

Displays the graphic that says “Not Enough Homes!” when the player tries to add more people if there’s not enough housing for them.

`void newTile()`

Displays the graphic that says “New Tile” when the player unlocks a new tile.

`void mustConnectResidences()`

Displays the graphic that says “Connect Homes By Roads” when the player tries to add people, there’s enough space for them, and some homes on the active tile are disconnected from roads.

`void NotEnoughPeople()`

Displays the graphic that says “NotEnoughPeople” when the player tries to build a workplace and there aren’t enough available people to be employees.

`void MaxCarbonCaptureTilesError()`

Displays the graphic that tells the player what the maximum number of carbon capture tiles per area is.

`void UseTrashButtonToRemoveTiles()`

Displays the error that says “Use To Delete Tiles” and points at the trash button, when the player tries to place a tile over another tile for the first time.

`void MustPurchaseAreaError()`

Displays the error that says “Must Purchase Area” when the player tries to place tiles on an un-purchased area.

`void NewAreaUnlockedNotification()`

Displays the graphic that says “New Area” when the first area unlocks.

ErrorGraphicDisplay.cs

The error graphic displayer class is in charge of turning notification graphics on when told and turning them off after a certain period of time. These are some of the class's important public functions:

```
void SetValues(float _timeToShowGraphic, GameObject _errorGraphic)
```

Updates how long the given error graphic should be displayed as well as what the graphic's game object is.

```
void DisplayErrorGraphic()
```

Activates the graphic associated with the given error graphic displayer.

-Graydon Bruyn, June 2026